Does a Pretrained Deep EEG Source-Imaging Model Agree with fMRI on a Visual Task?

A reproduction and cross-modal check on NATVIEW

Lintong Han | lintongbarry.github.io

Technical report, 2026. Independent work. Not peer reviewed.

Abstract

Deep-learning methods for EEG source imaging are usually validated on simulated data, where the true sources are known. I asked a simpler question about real data: when a published, pretrained model (DeepSIF) is applied to EEG recorded during a flickering-checkerboard task, does its source estimate land where simultaneously recorded fMRI shows activation? I reproduced DeepSIF on one subject of the open NATVIEW simultaneous EEG-fMRI dataset, projected both the EEG source estimate and an fMRI on–off difference map onto the same fsaverage5 cortical surface, and compared them at the global, regional and peak-overlap levels. The two maps did not agree. Their global correlation was near zero ($r = -0.012$), the overlap of their strongest 5% of vertices was at or below what random placement would give (Dice 0.040 against a chance level of 0.050), and neither map was strongly concentrated in posterior cortex. Because the fMRI reference itself was only weakly posterior, the disagreement cannot yet be attributed to the model. This report documents the pipeline and the negative result, lists the likely causes, and sets out the diagnostics I plan next.

1 Motivation

Electrophysiological source imaging (ESI) reconstructs cortical activity from scalp EEG. DeepSIF trains a deep network on data simulated with neural mass models and a realistic head model, and its authors report better spatiotemporal estimates than conventional ESI [1]. Most of this evidence comes from simulations and from clinical recordings in epilepsy. A healthy visual task with a simultaneous fMRI recording offers a different kind of check: we do not know the true sources, but we do know roughly where a checkerboard should drive activity (occipital visual cortex), and we have an independent measurement of where blood flow changed. I wanted to know whether the pretrained model, used off the shelf, passes this check.

2 Data

I used the NATVIEW dataset [2], which contains simultaneous EEG-fMRI from 22 healthy adults recorded at the Nathan S. Kline Institute. Among its tasks is a 200-second flickering checkerboard (12 Hz), recorded outside the scanner, inside the scanner with the scanner off, and during simultaneous EEG-fMRI. EEG was sampled at 5000 Hz. This report uses the simultaneous checkerboard run of subject sub-01.

3 Pipeline

3.1 EEG

- **Quality checks.** I inspected the amplitude distribution across all epochs and channels, the per-channel variance (to flag flat, noisy or outlier channels), and the power spectrum at occipital channels (Figure 1).

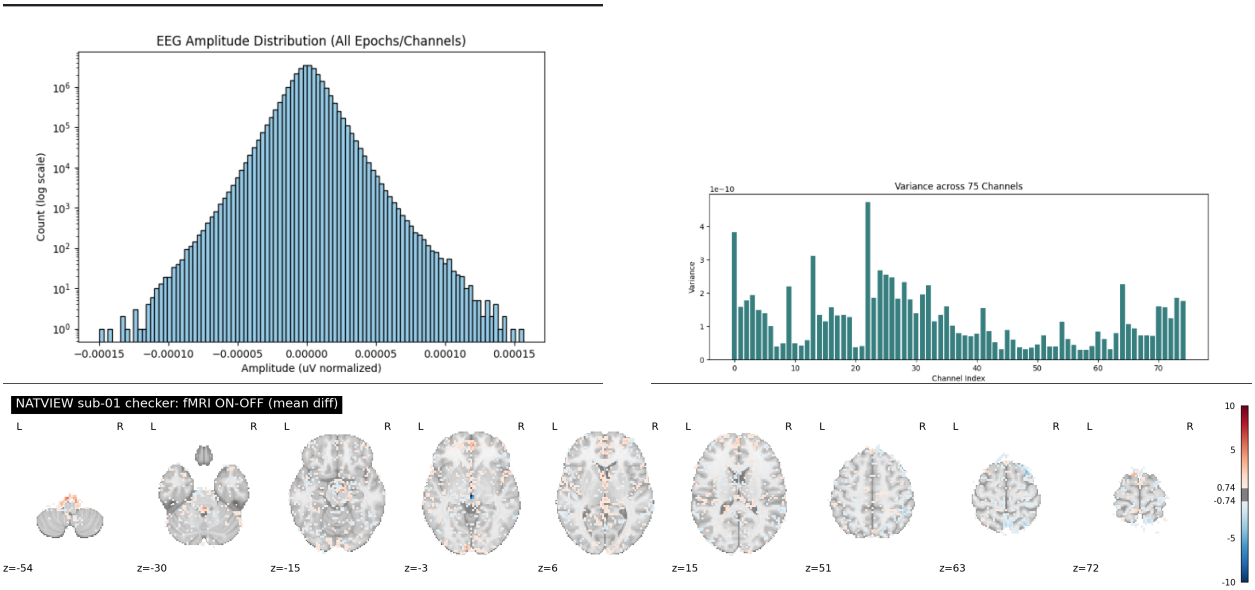


Figure 1: Quality checks. Top left: EEG amplitude distribution across all epochs and channels. Top right: variance per channel after alignment to the 75-channel layout. Bottom: axial slices of the fMRI on–off difference volume.

- **Channel alignment.** DeepSIF expects a fixed 75-channel input. I mapped the NATVIEW channels to that layout and ordering (file `X75`).
- **Model-matched preprocessing and inference.** I preprocessed the data to match the model's expected input and ran the authors' pretrained weights, which output activity over 994 cortical source regions through time.
- **Stimulus map.** From the inferred time courses I formed a stimulus-only source map (`s_map_stim`).

3.2 fMRI reference

I formed an on–off difference volume (mean signal during checkerboard blocks minus mean during rest blocks) and sampled it at the pial-surface vertices of fsaverage5. This is a deliberately simple reference; Section 5 explains why it is likely one of the weak points.

3.3 Common surface

Both maps were placed on the same fsaverage5 pial surface [3], with 10,242 vertices per hemisphere ($N = 20{,}484$ in total). I tried two projection routes and report the fsaverage5 route here.

3.4 Comparison metrics

- **Global:** Pearson correlation between the two vertex maps.
- **Peak overlap:** for $p \in \{0.5, 1, 2, 5\}\%$, take the top $k = pN$ vertices of each map by value and compute the size of their intersection, the Dice coefficient and the intersection-over-union (IoU).
- **Regional:** for posterior regions of interest defined by the y coordinate ($y < -30$, -40, -50 or -60 mm), the fraction of vertices inside the region and, for each map, the ratio of mean absolute

value inside the region to mean absolute value over the whole cortex. A ratio well above 1 means the map concentrates in posterior cortex.

For the peak-overlap metric, a useful reference is chance: if two sets of k vertices were placed at random among N, the expected intersection is k^2/N and the expected Dice is $k/N = p$.

4 Results

Table 1: Peak overlap between the EEG source map and the fMRI map on fsaverage5 ($N = 20{,}484$). Chance values assume random placement.

Top p	k	Intersection	Expected by chance	Dice	Chance Dice
0.5%	–	–	–	0.000	0.005
1%	205	0	2.1	0.000	0.010
2%	410	1	8.2	0.002	0.020
5%	1024	41	51.2	0.040	0.050

Global agreement. The Pearson correlation between the two maps was $r = -0.0116$, i.e. no linear relationship.

Peak overlap. At every threshold the overlap was at or below chance (Table 1). At the top 5%, the maps shared 41 vertices where random placement would give about 51. Restricting the comparison to posterior cortex ($y < -40$ mm) did not change the picture: top-5% Dice 0.047, IoU 0.024.

Table 2: Concentration in posterior cortex: mean absolute value inside the region divided by mean absolute value over the whole cortex.

Region	Fraction of vertices	EEG (DeepSIF) ratio	fMRI ratio
$y < -30$ mm	0.430	1.035	1.055
$y < -40$ mm	0.330	1.035	1.067
$y < -50$ mm	0.248	1.050	1.098
$y < -60$ mm	0.173	1.027	1.125

Regional concentration. Both ratios stayed close to 1 (Table 2). The fMRI ratio rose steadily toward the occipital pole, reaching 1.125 for $y < -60$ mm, which is the expected direction but a weak effect for a full-field checkerboard. The EEG ratio did not rise. Visually, the DeepSIF map consists of small, scattered peaks spread over the whole cortex, while the fMRI difference map shows broad, noisy positive and negative patches (Figure 2).

5 Why the maps might disagree

The result does not by itself show that DeepSIF fails on real visual data. Several parts of the pipeline could produce the same disagreement, and I have not yet separated them.

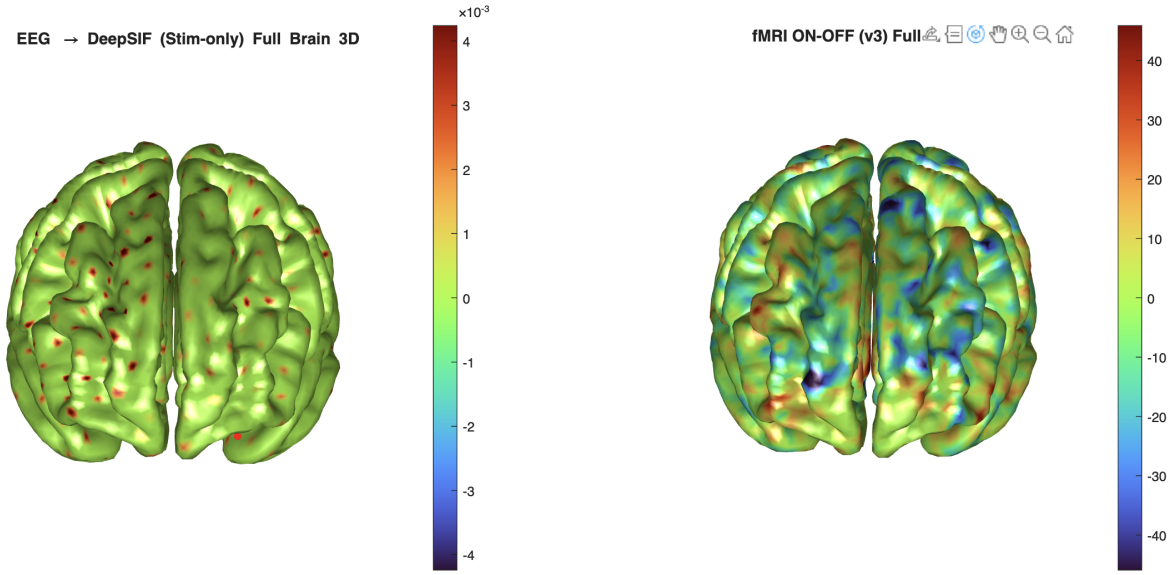


Figure 2: Left: stimulus-only DeepSIF source map. Right: fMRI on–off difference map. Both on the fsaverage5 surface, dorsal view.

1. **The fMRI reference is weak.** A raw on–off mean difference ignores the hemodynamic delay and does not model noise or motion. The modest posterior ratio of the fMRI map (Table 2) suggests the reference itself does not cleanly show visual cortex.
2. **The stimulus is a steady-state one.** A 12 Hz flicker drives a steady-state visual evoked response at the flicker frequency. Averaging source power over the whole stimulus period, rather than looking at the 12 Hz component, may wash out the visual response.
3. **In-scanner EEG artifacts.** Gradient and cardiac (ballistocardiogram) residuals in simultaneously recorded EEG are large compared with visual responses.
4. **Input mismatch with the model.** DeepSIF was trained with a particular head model, channel layout and signal scale. The amplitude axis in my quality check (Figure 1) suggests that units and scaling need to be verified against what the model expects. The channel mapping onto the 75-channel layout is another possible source of error.
5. **Template anatomy.** Both maps were placed on a template surface, not the subject's own cortex, which blurs the location of small visual areas.

6 Planned diagnostics

Each step below is chosen to rule causes in or out, not to make the maps agree.

1. **Sensor-level check first.** Confirm a 12 Hz peak at occipital channels, comparing the outside-scanner, scanner-off and simultaneous runs. If the peak is absent in the simultaneous run, source imaging cannot be expected to recover it.
2. **A proper fMRI reference.** Fit a general linear model with a block regressor convolved with a hemodynamic response function, produce a z-map, confirm activation in visual cortex, and project it to the surface through the subject's anatomical scan.

3. **A classical inverse baseline.** Run eLORETA or dSPM [4, 5] on the same data with a template head model, using the 12 Hz component. If the classical estimate finds occipital cortex and DeepSIF does not, the problem is model transfer; if neither does, it is the data or the preprocessing.
4. **A proper null.** Replace the analytic chance level with a spin test [6], which preserves the spatial smoothness of each map.
5. **More subjects.** Repeat on additional NATVIEW subjects before drawing any conclusion about the model.

7 Limitations

This is a single-subject analysis on template anatomy with a simple fMRI reference. I did not tune the model or the preprocessing to improve agreement, and the numbers above should be read as the outcome of an off-the-shelf reproduction, not as an evaluation of DeepSIF.

References

[1] R. Sun, A. Sohrabpour, G. A. Worrell, B. He. Deep neural networks constrained by neural mass models improve electrophysiological source imaging of spatiotemporal brain dynamics. *PNAS* 119(31):e2201128119, 2022.

[2] Q. K. Telesford et al. An open-access dataset of naturalistic viewing using simultaneous EEG-fMRI. *Scientific Data* 10:554, 2023.

[3] B. Fischl, M. I. Sereno, R. B. H. Tootell, A. M. Dale. High-resolution intersubject averaging and a coordinate system for the cortical surface. *Human Brain Mapping* 8(4):272–284, 1999.

[4] R. D. Pascual-Marqui. Discrete, 3D distributed, linear imaging methods of electric neuronal activity. Part 1: exact, zero error localization. arXiv:0710.3341, 2007.

[5] A. Gramfort et al. MEG and EEG data analysis with MNE-Python. *Frontiers in Neuroscience* 7:267, 2013.

[6] A. F. Alexander-Bloch et al. On testing for spatial correspondence between maps of human brain structure and function. *NeuroImage* 178:540–551, 2018.